

| Schedule d Week | Module Title                       |       |              | Learning Units                                                                                                                                                                                                                                                           | Remarks                                                                                           |
|-----------------|------------------------------------|-------|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Week 1          | Introduction                       | Day 1 | Hour# 1      | <ul style="list-style-type: none"> <li>• Introduction to AI</li> <li>• Motivational Lecture (For further detail please see Page No: 3&amp; 4)</li> </ul>                                                                                                                 | <ul style="list-style-type: none"> <li>• Task 1</li> <li>• Task 2</li> <li>• Task 3-25</li> </ul> |
|                 |                                    |       | Hour# 2      | <ul style="list-style-type: none"> <li>• Course Introduction</li> <li>• Job market</li> <li>• Course Applications</li> <li>• Work ethics</li> <li>• Survey of career opportunities</li> <li>• Survey of industry requirements for each career path</li> <li>•</li> </ul> |                                                                                                   |
|                 |                                    |       | Hour# 3, 4   | •Software Installation (Anaconda, VSCode, PyCharm, etc.)                                                                                                                                                                                                                 |                                                                                                   |
|                 | Linux Shell Scripting Fundamentals | Day 2 | Hour# 1      | Introduction to Debian<br>•Basic Commands: pwd, cd, ls, cat, sudo, man, redirection, mkdir, rm, rmdir, cp, mv                                                                                                                                                            |                                                                                                   |
|                 |                                    |       | Hour# 2      | •file, reading, cat, more, less, head, alias,                                                                                                                                                                                                                            |                                                                                                   |
|                 |                                    |       | Hour# 3      | •shutdown, restart, touch, nano, bash, sh, chmod, ps, kill, dpkg                                                                                                                                                                                                         |                                                                                                   |
|                 |                                    |       | Hour# 4      | •Package update and upgrade•Environment                                                                                                                                                                                                                                  |                                                                                                   |
|                 | Python Fundamentals                | Day 3 | Hour # 1     | Values, expressions, and statements<br>•Numbers, Booleans, Strings<br>•Operators, variables and keywords                                                                                                                                                                 | <u>Details may be seen at Annexure-I</u>                                                          |
|                 |                                    |       | Hour # 2,3   | •String operations                                                                                                                                                                                                                                                       |                                                                                                   |
|                 |                                    |       | Hour # 4     | •Input and Type casting<br>•Comments                                                                                                                                                                                                                                     |                                                                                                   |
|                 |                                    | Day 4 | Hour # 1 & 2 | Data Structures<br>• Lists<br>• Tuples                                                                                                                                                                                                                                   |                                                                                                   |
|                 |                                    |       | Hour # 3 & 4 | • Dictionaries<br>• Sets                                                                                                                                                                                                                                                 |                                                                                                   |
|                 |                                    | Day 5 | Hour # 1 & 2 | Conditional Execution<br>•If, elif, and else statements<br>•Break, continue, and pass statements                                                                                                                                                                         |                                                                                                   |
|                 |                                    |       |              |                                                                                                                                                                                                                                                                          |                                                                                                   |

|               |                                            |       |                |                                                                                                                                                                                                           |                                                                                                                                      |
|---------------|--------------------------------------------|-------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
|               |                                            |       |                | <ul style="list-style-type: none"> <li>•Nested conditionals</li> <li>•Conditional (Ternary) Expression</li> </ul>                                                                                         |                                                                                                                                      |
|               |                                            |       | Hour#<br>3 & 4 | <ul style="list-style-type: none"> <li>•While, for loops and use of enumerate</li> <li>•Nested loops</li> <li>•List comprehension</li> <li>•Iterators and Iterables</li> <li>•</li> </ul>                 |                                                                                                                                      |
| <b>Week 2</b> | Python Fundamentals                        | Day 1 | Hour#<br>1     | <ul style="list-style-type: none"> <li>• Motivational Lecture (For further detail please see Page No: 3&amp; 4)</li> </ul>                                                                                | <ul style="list-style-type: none"> <li>• Task 26-27</li> <li>• Task 49-51</li> </ul> <p><u>Details may be seen at Annexure-I</u></p> |
|               | Implementation of OOP Principals in Python |       | Hour#<br>2, 3  | <ul style="list-style-type: none"> <li>• Functions</li> <li>• Functions and variable scope</li> <li>• Lambda expression</li> <li>• Map and Filter</li> <li>• Inner/Nested functions</li> </ul>            |                                                                                                                                      |
|               |                                            |       | Hour#<br>4     | <ul style="list-style-type: none"> <li>• File Handling</li> <li>• Exception Handling</li> </ul>                                                                                                           |                                                                                                                                      |
|               |                                            | Day 2 | Hour#<br>1     | <ul style="list-style-type: none"> <li>• Classes and Objects</li> <li>• Instance Variables and Methods</li> <li>• Class Variables and Functions</li> <li>• Constructors and Destructors</li> </ul>        |                                                                                                                                      |
|               | Descriptive Statistics and Probability     |       | Hour#<br>2,3   | <ul style="list-style-type: none"> <li>• Inheritance</li> <li>• Multilevel Inheritance</li> <li>• Hierarchical Inheritance</li> <li>• Multiple Inheritance, Method Resolution Order</li> </ul>            |                                                                                                                                      |
|               |                                            |       | Hour#<br>4     | <ul style="list-style-type: none"> <li>• Access Specifiers: Private, Public, Protected</li> <li>• Name Mangling</li> <li>• Inner/Nested Class</li> <li>• Association, Aggregation, Composition</li> </ul> |                                                                                                                                      |
|               |                                            | Day 3 | Hour#<br>1     | <ul style="list-style-type: none"> <li>• Polymorphism and Operator Overloading</li> </ul>                                                                                                                 |                                                                                                                                      |
|               |                                            |       | Hour#<br>2     | <ul style="list-style-type: none"> <li>• Magic Functions/Dunder Functions</li> </ul>                                                                                                                      |                                                                                                                                      |
|               |                                            |       | Hour#<br>3     | <ul style="list-style-type: none"> <li>• Dynamic Polymorphism (subclass as base class)</li> </ul>                                                                                                         |                                                                                                                                      |
|               |                                            |       | Hour#<br>4     | <ul style="list-style-type: none"> <li>• Abstract Method and Class, Empty Class, Data Class</li> <li>• Keyword Arguments</li> </ul>                                                                       |                                                                                                                                      |
|               |                                            | Day 4 | Hour#<br>1, 2  | <ul style="list-style-type: none"> <li>• Data and its types (structured, Unstructured)</li> <li>• Quantitative data, Numerical, Continuous, and Discrete variables</li> </ul>                             |                                                                                                                                      |

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|               | Overview                                                          |       |             | <ul style="list-style-type: none"> <li>Qualitative data, Categorical, Nominal, Ordinal, and Binary variables</li> </ul>                                                                                                                                                                                                                                       |                                                              |
|               |                                                                   |       | Hour # 3-4  | Measures of Central Tendency <ul style="list-style-type: none"> <li>Mean, Mode, Median</li> </ul>                                                                                                                                                                                                                                                             |                                                              |
|               |                                                                   | Day 5 | Hour# 1,2   | Measures of Dispersion <ul style="list-style-type: none"> <li>Variance, Standard deviation</li> <li>Co-efficient of variation, skewness and kurtosis</li> </ul>                                                                                                                                                                                               |                                                              |
|               |                                                                   |       | Hour# 3, 4  | Measures of Position <ul style="list-style-type: none"> <li>Z-Score, Percentile, Quartile</li> </ul>                                                                                                                                                                                                                                                          |                                                              |
| <b>Week 3</b> | Descriptive Statistics and Probability Overview                   | Day 1 | Hour# 1     | <ul style="list-style-type: none"> <li>Motivational Lecture (For further detail please see Page No: 3&amp; 4)</li> </ul>                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Task 28-48</li> </ul> |
|               |                                                                   |       | Hour# 2     | <ul style="list-style-type: none"> <li>Correlation Coefficient</li> </ul>                                                                                                                                                                                                                                                                                     |                                                              |
|               |                                                                   |       | Hour# 3     | <ul style="list-style-type: none"> <li>Univariate, bivariate and multivariate plots</li> </ul>                                                                                                                                                                                                                                                                |                                                              |
|               |                                                                   |       | Hour# 4     | Probability                                                                                                                                                                                                                                                                                                                                                   |                                                              |
|               | Python Support Libraries for Exploratory Data Analysis<br>- NUMPY | Day 2 | Hour# 1     | Joint, Marginal and Conditional probability                                                                                                                                                                                                                                                                                                                   | <u>Details may be seen at Annexure-I</u>                     |
|               |                                                                   |       | Hour# 2     | <ul style="list-style-type: none"> <li>Probability Distributions</li> </ul>                                                                                                                                                                                                                                                                                   |                                                              |
|               |                                                                   |       | Hour # 3-4  | <ul style="list-style-type: none"> <li>Discrete and Continuous probability distributions</li> <li>Bayesian Probability</li> </ul>                                                                                                                                                                                                                             |                                                              |
|               |                                                                   | Day 3 | Hour# 1     | <ul style="list-style-type: none"> <li>Introduction to Numpy</li> </ul>                                                                                                                                                                                                                                                                                       |                                                              |
|               |                                                                   |       | Hour# 2,3,4 | <ul style="list-style-type: none"> <li>Creating Numpy Arrays (from Python list, from built-in methods, from random)</li> <li>Array Attributes and Methods (reshape, max, min, argmax, argmin, shape, dtype, size, ndim)</li> <li>Operations on Arrays (copying, append and Insert, Sorting, Removing/Deleting, Combining/Concatenating, Splitting)</li> </ul> |                                                              |
|               |                                                                   | Day 4 | Hour # 1-2  | <ul style="list-style-type: none"> <li>Data Loading &amp; Saving</li> <li>NumPy Indexing and Selection (Indexing a 2D array, Logical Selection)</li> <li>Broadcasting</li> </ul>                                                                                                                                                                              |                                                              |
|               |                                                                   |       | Hour # 3-4  | <ul style="list-style-type: none"> <li>Type Casting</li> <li>Arithmetic Operations (Add, Subtract, Multiply, Divide, Exponentiation)</li> <li>Universal Array Functions (sqrt, exp, max,</li> </ul>                                                                                                                                                           |                                                              |

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|               |                                                                                 |       |             | sin, etc)                                                                                                                                                                                                     |                                                          |
|               | - Pandas                                                                        | Day 5 | Hour# 1     | • Introduction to Pandas                                                                                                                                                                                      |                                                          |
|               |                                                                                 |       | Hour# 2     | • Series and DataFrame and Data Input<br>• Selection and Indexing (rows, columns, conditional selection, selection of subset of rows and columns, index setting, etc)                                         |                                                          |
|               |                                                                                 |       | Hour# 3     | • Operations on DataFrames (head, unique, value counts, applying custom functions, getting column and index names, sorting and ordering, null value check, value replacement, dropping rows and columns, etc) |                                                          |
|               |                                                                                 |       | Hour# 4     | • Missing data & its handling                                                                                                                                                                                 |                                                          |
| <b>Week 4</b> | Python Support Libraries for Exploratory Data Analysis<br>- Pandas<br>- Seaborn | Day 1 | Hour# 1     | • Motivational Lecture (For further detail please see Page No: 3& 4)                                                                                                                                          |                                                          |
|               |                                                                                 |       | Hour# 2     | • Merging, Joining, and Concatenation (inner, outer, right and left joins)                                                                                                                                    |                                                          |
|               |                                                                                 |       | Hour # 3-4  | • GroupBy<br>Discretization and Binning<br>• Operations on DataFrames<br>Data output/saving<br>• Pandas for Plotting (area, bar, density, hist, line, scatter, barh, box, hexbin, kde, and pie plots)         | • Task 28-48<br><u>Details may be seen at Annexure-I</u> |
|               |                                                                                 | Day 2 | Hour# 1     | • Introduction to Seaborn                                                                                                                                                                                     |                                                          |
|               |                                                                                 |       | Hour# 2     | Distribution Plots<br>• distplot<br>• jointplot (pairplot, rugplot, kdeplot)                                                                                                                                  |                                                          |
|               |                                                                                 |       | Hour# 3     | Categorical Data Plots<br>• factorplot, boxplot, violinplot, stripplot, swarmplot, barplot, countplot                                                                                                         |                                                          |
|               |                                                                                 |       | Hour# 4     | Matrix Plots<br>• Heatmap                                                                                                                                                                                     |                                                          |
|               |                                                                                 | Day 3 | Hour# 1     | • Machine learning introduction and types                                                                                                                                                                     |                                                          |
|               |                                                                                 |       | Hour# 2,3,4 | • Classical machine learning pipeline (data collection, preprocessing, feature crafting, modeling, testing and evaluation, and deployment)                                                                    |                                                          |
|               |                                                                                 | Day 4 | Hour # 1,2  | • Supervised machine learning<br>Regression and classification problems<br>• Components of supervised machine learning (labeled data, hypothesis, cost function, optimizer)                                   |                                                          |
|               |                                                                                 |       | Hour # 3,4  | • Univariate Linear Regression with Gradient Descent                                                                                                                                                          |                                                          |

|               |                             |       |                |                                                                                                                                                                                                          |                                                                                                                |
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|               |                             | Day 5 | Hour # 1-2     | <ul style="list-style-type: none"> <li>Univariate Linear Regression with Gradient Descent</li> <li>Without Vectorization</li> </ul>                                                                      |                                                                                                                |
|               |                             |       | Hour # 3-4     | <ul style="list-style-type: none"> <li>With Vectorization</li> </ul>                                                                                                                                     |                                                                                                                |
| <b>Week 5</b> | Machine Learning-I          | Day 1 | Hour# 1        | <ul style="list-style-type: none"> <li>Motivational Lecture (For further detail please see Page No: 3&amp; 4)</li> </ul>                                                                                 | <ul style="list-style-type: none"> <li>Task – 51,52</li> </ul> <p><u>Details may be seen at Annexure-I</u></p> |
|               |                             |       | Hour# 2,3,4    | <ul style="list-style-type: none"> <li>Multivariate Linear Regression</li> </ul>                                                                                                                         |                                                                                                                |
|               |                             | Day 2 | Hour# 1,2,3, 4 | <ul style="list-style-type: none"> <li>Polynomial Regression</li> </ul>                                                                                                                                  |                                                                                                                |
|               |                             | Day 3 | Hour# 1,2,3, 4 | <ul style="list-style-type: none"> <li>Logistic Regression (Binary Classification)</li> </ul>                                                                                                            |                                                                                                                |
|               |                             | Day 4 | Hour# 1,2,3, 4 | <ul style="list-style-type: none"> <li>Logistic Regression (Multiclass Classification)</li> </ul>                                                                                                        |                                                                                                                |
|               |                             | Day 5 | Hour# 1,2,3, 4 | <ul style="list-style-type: none"> <li>Code practice</li> </ul>                                                                                                                                          |                                                                                                                |
| <b>Week 6</b> | Natural Language Processing | Day 1 | Hour# 1        | <ul style="list-style-type: none"> <li>Motivational Lecture (For further detail please see Page No: 3&amp; 4)</li> </ul>                                                                                 | <ul style="list-style-type: none"> <li>Task 53-55</li> </ul> <p><u>Details may be seen at Annexure-I</u></p>   |
|               |                             |       | Hour# 2        | <ul style="list-style-type: none"> <li>Introduction to Natural Language Processing</li> </ul>                                                                                                            |                                                                                                                |
|               |                             |       | Hour# 3        | <ul style="list-style-type: none"> <li>Syntax, Semantics, Pragmatics, and Discourse</li> <li>NLP curves and future directions</li> </ul>                                                                 |                                                                                                                |
|               |                             |       | Hour# 4        | Data pre-processing for NLP <ul style="list-style-type: none"> <li>Introduction to NLTK/SpaCy</li> <li>Noise removal (stopwords, punctuation, etc)</li> </ul>                                            |                                                                                                                |
|               |                             | Day 2 | Hour# 1        | <ul style="list-style-type: none"> <li>Word and sentence tokenization</li> <li>Word segmentation</li> <li>Stemming</li> <li>Text normalization</li> <li>Regular expression for string parsing</li> </ul> |                                                                                                                |
|               |                             |       |                | <ul style="list-style-type: none"> <li>POS tagging</li> <li>NER tagging</li> <li>Chunking and Chinking</li> <li>Lemmatization</li> <li>WordNet</li> </ul>                                                |                                                                                                                |
|               |                             |       | Hour# 4        | <ul style="list-style-type: none"> <li>Words as features (BoW model)</li> <li>Feature Selection and Extraction</li> <li>Document Similarity</li> </ul>                                                   |                                                                                                                |
|               |                             |       |                |                                                                                                                                                                                                          |                                                                                                                |
|               | Machine Learning II         | Day 3 | Hour# 1        | <ul style="list-style-type: none"> <li>Testing</li> </ul>                                                                                                                                                |                                                                                                                |
|               |                             |       | Hour# 2        | <ul style="list-style-type: none"> <li>Evaluation Metrics</li> <li>Classification and Regression</li> </ul>                                                                                              |                                                                                                                |

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|                                                            |                  |       | Hour # 3-4     | <ul style="list-style-type: none"> <li>Dataset imbalance and its remedies (Augmentation)</li> </ul>                                                                                                                                                                                                                                     |                                                                                                              |
|                                                            |                  | Day 4 | Hour# 1,2,3    | <ul style="list-style-type: none"> <li>Support Vector Machine (SVM)</li> </ul>                                                                                                                                                                                                                                                          |                                                                                                              |
|                                                            |                  |       | Hour# 4        | <ul style="list-style-type: none"> <li>Decision Tree</li> </ul>                                                                                                                                                                                                                                                                         |                                                                                                              |
|                                                            |                  | Day 5 | Hour# 1,2      | <ul style="list-style-type: none"> <li>Decision Tree</li> </ul>                                                                                                                                                                                                                                                                         |                                                                                                              |
|                                                            |                  |       | Hour # 3-4     | <ul style="list-style-type: none"> <li>Bagging – Random Forest</li> </ul>                                                                                                                                                                                                                                                               |                                                                                                              |
| <p align="center"><b>Build Your CV – Mid-term Exam</b></p> |                  |       |                |                                                                                                                                                                                                                                                                                                                                         |                                                                                                              |
| <b>Week 7</b>                                              | Deep Learning I  | Day 1 | Hour# 1        | <ul style="list-style-type: none"> <li>Motivational Lecture (For further detail please see Page No: 3&amp; 4)</li> </ul>                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Task 56-64</li> </ul> <p><u>Details may be seen at Annexure-I</u></p> |
|                                                            |                  |       | Hour# 2,3,4    | <ul style="list-style-type: none"> <li>Boosting</li> </ul>                                                                                                                                                                                                                                                                              |                                                                                                              |
|                                                            |                  | Day 2 | Hour# 1,2,3, 4 | MLP Feed Forward Neural Network <ul style="list-style-type: none"> <li>Forward and backward passes</li> <li>Nonlinearity: Activation functions</li> <li>Cross-Entropy</li> <li>Computational graph and Backpropagation</li> <li>Vanishing and exploding gradients</li> <li>Overfitting, underfitting, dropout regularization</li> </ul> |                                                                                                              |
|                                                            |                  | Day 3 | Hour# 1,2,3, 4 | <ul style="list-style-type: none"> <li>Introduction and implementation of neural networks using appropriate deep learning API of choice (TensorFlow, PyTorch, Keras)</li> </ul>                                                                                                                                                         |                                                                                                              |
|                                                            |                  | Day 4 | Hour # 1-2     | Convolutional Neural Network (CNN) <ul style="list-style-type: none"> <li>2D CNN for image classification</li> </ul>                                                                                                                                                                                                                    |                                                                                                              |
|                                                            |                  |       | Hour # 3-4     | <ul style="list-style-type: none"> <li>1D CNN for text document classification</li> </ul>                                                                                                                                                                                                                                               |                                                                                                              |
|                                                            |                  | Day 5 | Hour # 1-2     | <ul style="list-style-type: none"> <li>Code Practice Neural Networks</li> </ul>                                                                                                                                                                                                                                                         |                                                                                                              |
|                                                            |                  |       | Hour # 3-4     | <ul style="list-style-type: none"> <li>Code Practice Neural Networks</li> </ul>                                                                                                                                                                                                                                                         |                                                                                                              |
| <b>Week 8</b>                                              | Deep Learning II | Day 1 | Hour# 1        | <ul style="list-style-type: none"> <li>Motivational Lecture (For further detail please see Page No: 3&amp; 4)</li> </ul>                                                                                                                                                                                                                |                                                                                                              |
|                                                            |                  |       | Hour# 2,3,4    | <ul style="list-style-type: none"> <li>Recurrent Neural Networks (RNNs)</li> </ul>                                                                                                                                                                                                                                                      |                                                                                                              |
|                                                            |                  | Day 2 | Hour# 1,2,3, 4 | <ul style="list-style-type: none"> <li>Long-Short-Term-Memory Networks (LSTM)</li> </ul>                                                                                                                                                                                                                                                |                                                                                                              |
|                                                            |                  | Day 3 | Hour# 1        | <ul style="list-style-type: none"> <li>LSTM Code Practice</li> </ul>                                                                                                                                                                                                                                                                    |                                                                                                              |
|                                                            |                  | Day 4 | Hour# 1,2,3, 4 | <ul style="list-style-type: none"> <li>Gated Recurrent Unit Networks</li> </ul>                                                                                                                                                                                                                                                         |                                                                                                              |

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|                |                                                                                                                     | Day 5   | Hour#1,2,3,4 | <ul style="list-style-type: none"> <li>GRU Code Practice</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                           |
| <b>Week 9</b>  | Deep Learning II                                                                                                    | Day 1   | Hour#1       | <ul style="list-style-type: none"> <li>Motivational Lecture (For further detail please see Page No: 3&amp; 4)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                           |
|                |                                                                                                                     |         | Hour#2,3,4   | Word Embeddings <ul style="list-style-type: none"> <li>Word2vec</li> <li>Continuous BOW</li> <li>Continuous Skip-gram</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                           |
|                |                                                                                                                     | Day 2   | Hour#1,2,3,4 | <ul style="list-style-type: none"> <li>Gensim and Custom Embedding Training</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                           |
|                |                                                                                                                     | Day 3   | Hour#1,2,3,4 | <ul style="list-style-type: none"> <li>Sequence Models</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
|                |                                                                                                                     | Day 4   | Hour#1,2,3,4 | Sequence Models <ul style="list-style-type: none"> <li>1 to 1</li> <li>1 to Many</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                           |
|                |                                                                                                                     | Day 5   | Hour#1,2,3,4 | Sequence Models <ul style="list-style-type: none"> <li>Many to 1</li> <li>Many to Many</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
| <b>Week 10</b> | Deep Learning II                                                                                                    | Day 1   | Hour#1       | <ul style="list-style-type: none"> <li>Motivational Lecture (For further detail please see Page No: 3&amp; 4)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <ul style="list-style-type: none"> <li>Task 65</li> </ul> <p><u>Details may be seen at Annexure-I</u></p> |
|                |                                                                                                                     |         | Hour#2,3,4   | <ul style="list-style-type: none"> <li>Bi-Directional LSTM/RNN in Sequence Models</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                           |
|                |                                                                                                                     | Day 2,3 | Hour#1,2,3,4 | <ul style="list-style-type: none"> <li>Attention Mechanism in Models</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                           |
|                |                                                                                                                     | Day 4,5 | Hour#1,2,3,4 | Selection of Project, architecture discussion, preparation. <ul style="list-style-type: none"> <li>Guidelines to the Trainees for selection of employable project like final year project (FYP).</li> <li>Assignment of Independent project to each Trainee.</li> <li>A project based on trainee's aptitude and acquired skills.</li> <li>Designed by keeping in view the emerging trends in the local market as well as across the globe.</li> <li>The project idea may be based on entrepreneurship.</li> <li>Leading to the successful employment.</li> <li>The duration of the project will be 2 weeks Ideas may be generated via different sites such as:<br/> <a href="https://1000projects.org/">https://1000projects.org/</a><br/> <a href="https://nevonprojects.com/">https://nevonprojects.com/</a><br/> <a href="https://www.freestudentprojects.com/">https://www.freestudentprojects.com/</a> </li> </ul> |                                                                                                           |
|                | Employable Project / Assignment (2 weeks, 11-12) in addition of regular classes.<br>OR<br>On job training (2 weeks) |         |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                           |

|                |                     |       |            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                           |
|----------------|---------------------|-------|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
|                |                     |       |            | <p><a href="https://technofizi.net/best-computer-science-and-engineering-cse-project-topics-ideas-for-students/">https://technofizi.net/best-computer-science-and-engineering-cse-project-topics-ideas-for-students/</a></p> <ul style="list-style-type: none"> <li>Final viva/assessment will be conducted on project assignments.</li> <li>At the end of session, the project will be presented in skills competition.</li> <li>The skill competition will be conducted on zonal, regional and National level.</li> <li>The project will be presented in front of Industrialists for commercialization</li> <li>The best business idea will be placed in NAVTTC business incubation center for commercialization.</li> </ul> <p><b>OR</b></p> <ul style="list-style-type: none"> <li>On job training for 2 weeks:</li> <li>Aims to provide 2 weeks industrial training to the Trainees as part of overall training program</li> <li>Ideal for the manufacturing trades</li> <li>As an alternate to the projects that involve expensive equipment</li> <li>Focuses on increasing Trainee's motivation, productivity, efficiency and quick learning approach.</li> </ul> |                                                                                                           |
| <b>Week 11</b> | MS Azure AI Service | Day 1 | Hour# 1    | <ul style="list-style-type: none"> <li>Motivational Lecture (For further detail please see Page No: 3&amp; 4)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <ul style="list-style-type: none"> <li>Task 65</li> </ul> <p><u>Details may be seen at Annexure-I</u></p> |
|                |                     |       | Hour # 2,3 | <ul style="list-style-type: none"> <li>Selection of Microsoft Azure AI Service</li> <li>Selection the appropriate service for a vision solution</li> <li>Selection the appropriate service for a language analysis solution</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                           |
|                |                     |       | Hour# 4    | <ul style="list-style-type: none"> <li>Selection the appropriate service for a decision support solution</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                           |
|                |                     | Day 2 | Hour # 1,2 | <ul style="list-style-type: none"> <li>Selection the appropriate service in Cognitive Services for a speech solution</li> <li>Selection the appropriate Applied AI services</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                           |
|                |                     |       | Hour # 3-4 | <ul style="list-style-type: none"> <li>Configuring Security for Microsoft Azure AI Service</li> <li>Manage account keys</li> <li>Manage authentication for a resource</li> <li>Secure services by using Azure Virtual Networks</li> <li>Plan for a solution that meets Responsible AI principles</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                           |
|                |                     | Day 3 | Hour # 1-2 | <ul style="list-style-type: none"> <li>Create &amp; Manage Microsoft Azure AI Service</li> <li>Create an Azure AI resource</li> <li>Configure diagnostic logging</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                           |

|                |  |       |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                             |
|----------------|--|-------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
|                |  |       | Hour # 3-4     | <ul style="list-style-type: none"> <li>• Manage costs for Azure AI services</li> <li>• Monitor an Azure AI resource</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                             |
|                |  | Day 4 | Hour # 1-2     | <ul style="list-style-type: none"> <li>• Deploy Microsoft Azure AI Services</li> <li>• Determine a default endpoint for a service</li> <li>• Create a resource by using the Azure portal</li> <li>• Integrate Azure AI services into a continuous integration/continuous deployment (CI/CD) pipeline</li> <li>• Plan a container deployment</li> <li>• Implement prebuilt containers in a connected environment</li> </ul>                                                                                                                                          |                                                                                                             |
|                |  |       | Hour # 3-4     | <p>Microsoft Azure Creation of Solutions for Anomaly Detection Content Improvement</p> <ul style="list-style-type: none"> <li>• Create a solution that uses Anomaly Detector, part of Cognitive Services</li> <li>• Create a solution that uses Azure Content Moderator</li> <li>• Create a solution that uses Personalizer, part of Cognitive Services</li> <li>• Create a solution that uses Azure Metrics Advisor, part of Azure Applied AI Services</li> <li>• Create a solution that uses Azure Immersive Reader, part of Azure Applied AI Services</li> </ul> |                                                                                                             |
|                |  | Day 5 | Hour # 1-2     | <p>Microsoft Azure Implementation of Image and Video Processing Solutions</p> <ul style="list-style-type: none"> <li>• Analyze images</li> <li>• Extract text from images</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                             |
|                |  |       | Hour # 3-4     | <ul style="list-style-type: none"> <li>• Implement image classification and object detection by using the Custom Vision service, part of Azure Cognitive Services</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                             |
|                |  |       |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                             |
| <b>Week 12</b> |  | Day 1 | Hour# 1        | <ul style="list-style-type: none"> <li>• Motivational Lecture (For further detail please see Page No: 3&amp; 4)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>• Task 65</li> </ul> <p><u>Details may be seen at Annexure-I</u></p> |
|                |  |       | Hour# 2,3,4    | <ul style="list-style-type: none"> <li>• Process videos</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                             |
|                |  | Day 2 | Hour# 1,2,3, 4 | <p>Microsoft Azure Natural Language Processing (NLP) Solutions Implementation</p> <ul style="list-style-type: none"> <li>• Analyze text</li> <li>• Process speech</li> <li>• Translate language</li> </ul>                                                                                                                                                                                                                                                                                                                                                          |                                                                                                             |
|                |  | Day 3 | Hour# 1,2,3, 4 | <ul style="list-style-type: none"> <li>• Build and manage a language understanding model</li> <li>• Create a question answering solution</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                             |
|                |  | Day 4 | Hour# 1        | <ul style="list-style-type: none"> <li>• Build and manage a language understanding model</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                             |
|                |  |       | Hour # 2-4     | <p>Microsoft Azure Knowledge Mining Solutions Implementation</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                             |
|                |  | Day 5 | Hour # 1-4     | <p>Microsoft Azure Conversational AI Solutions Implementation</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                             |



## Annexure-I

### List of Tasks

| Task No. | Task Title     | Description                                                                                                                                                                                                  | Week |
|----------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1.       | Installation   | Download and install Anaconda3<br>Install PyTorch<br>Install TensorFlow 2.0<br>Install VSCode<br>Install PyCharm                                                                                             | 1    |
| 2.       | Linux Commands | Practice these commands:<br>pwd, cd, ls, cat, sudo, man, redirection, mkdir, rm, rmdir, cp, mv, file, reading, cat, more, less, head, alias, shutdown, restart, touch, nano, bash, sh, chmod, ps, kill, dpkg | 1    |
| 3.       | Python         | <pre># This program adds two numbers  num1 = 1.5 num2 = 6.3  # Add two numbers sum = num1 + num2  # Display the sum print('The sum of {0} and {1} is {2}'.format(num1, num2, sum))</pre>                     | 1    |

|    |        |                                                                                                                                                                                                                                                                                   |   |
|----|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 4. | Python | <pre> # Store input numbers num1 = input('Enter first number: ') num2 = input('Enter second number: ')  # Add two numbers sum = float(num1) + float(num2)  # Display the sum print('The sum of {0} and {1} is {2}'.format(num1, num2, sum)) </pre>                                | 1 |
| 5. | Python | <pre> # Python Program to calculate the square root  # Note: change this value for a different result num = 8  # To take the input from the user #num = float(input('Enter a number: '))  num_sqrt = num ** 0.5 print('The square root of %0.3f is %0.3f'%(num ,num_sqrt)) </pre> | 1 |

|    |        |                                                                                                                                                                                                                                                                                                                          |   |
|----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 6. | Python | <pre> # Find square root of real or complex numbers # Importing the complex math module import cmath  num = 1+2j  # To take input from the user #num = eval(input('Enter a number: '))  num_sqrt = cmath.sqrt(num) print('The square root of {0} is {1:0.3f}+{2:0.3f}j'.format(num ,num_sqrt.real,num_sqrt.imag)) </pre> | 1 |
| 7. | Python | <pre> # Python Program to convert temperature in celsius to fahrenheit  # change this value for a different result celsius = 37.5  # calculate fahrenheit fahrenheit = (celsius * 1.8) + 32 print('%0.1f degree Celsius is equal to %0.1f degree Fahrenheit' %(celsius,fahrenheit)) </pre>                               | 1 |

|    |        |                                                                                                                                                                                                                                                                                                                                                                                                                        |   |
|----|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 8. | Python | <pre># Python Program to find the area of triangle  a = 5 b = 6 c = 7  # Uncomment below to take inputs from the user # a = float(input('Enter first side: ')) # b = float(input('Enter second side: ')) # c = float(input('Enter third side: '))  # calculate the semi-perimeter s = (a + b + c) / 2  # calculate the area area = (s*(s-a)*(s-b)*(s-c)) ** 0.5 print('The area of the triangle is %0.2f' %area)</pre> | 1 |
|----|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|

|     |        |                                                                                                                                                                                                                                                                                                                                |   |
|-----|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 9.  | Python | <pre> # Solve the quadratic equation ax**2 + bx + c = 0  # import complex math module import cmath  a = 1 b = 5 c = 6  # calculate the discriminant d = (b**2) - (4*a*c)  # find two solutions sol1 = (-b-cmath.sqrt(d))/(2*a) sol2 = (-b+cmath.sqrt(d))/(2*a)  print('The solution are {0} and {1}'.format(sol1,sol2)) </pre> | 1 |
| 10. | Python | <pre> # Taking kilometers input from the user kilometers = float(input("Enter value in kilometers: "))  # conversion factor conv_fac = 0.621371  # calculate miles miles = kilometers * conv_fac </pre>                                                                                                                        | 1 |
| 11. | Python | <pre> i = 10 if (i &gt; 15):     print ("10 is less than 15") print ("I am Not in if") </pre>                                                                                                                                                                                                                                  | 1 |

|     |        |                                                                                                                                                                                                                                                                                                                                   |   |
|-----|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 12. | Python | <pre> i = 20; if (i &lt; 15):     print ("i is smaller than 15")     print ("i'm in if Block") else:     print ("i is greater than 15")     print ("i'm in else Block") print ("i'm not in if and not in else Block") </pre>                                                                                                      | 1 |
| 13. | Python | <pre> i = 10 if (i == 10):     # First if statement     if (i &lt; 15):         print ("i is smaller than 15")     # Nested - if statement     # Will only be executed if statement above     # it is true     if (i &lt; 12):         print ("i is smaller than 12 too")     else:         print ("i is greater than 15") </pre> | 1 |
| 14. | Python | <pre> i = 20 if (i == 10):     print ("i is 10") elif (i == 15):     print ("i is 15") elif (i == 20):     print ("i is 20") else:     print ("i is not present") </pre>                                                                                                                                                          | 1 |
| 15. | Python | Exercise on for loops in Python:<br><a href="https://www.geeksforgeeks.org/python-for-loops/">https://www.geeksforgeeks.org/python-for-loops/</a>                                                                                                                                                                                 | 1 |
| 16. | Python | Exercise on While loops in Python:<br><a href="https://www.geeksforgeeks.org/python-while-loops/">https://www.geeksforgeeks.org/python-while-loops/</a>                                                                                                                                                                           | 1 |
| 17. | Python | Exercise on Break statement in Python:<br><a href="https://www.geeksforgeeks.org/python-break-statement/">https://www.geeksforgeeks.org/python-break-statement/</a>                                                                                                                                                               | 1 |
| 18. | Python | Exercise on Continue statement in Python:<br><a href="https://www.geeksforgeeks.org/python-continue-statement/">https://www.geeksforgeeks.org/python-continue-statement/</a>                                                                                                                                                      | 1 |
| 19. | Python | Exercise on various looping techniques in Python:<br><a href="https://www.geeksforgeeks.org/looping-techniques-python/">https://www.geeksforgeeks.org/looping-techniques-python/</a>                                                                                                                                              | 1 |
| 20. | Python | Exercise on User defined functions in Python:<br><a href="https://www.geeksforgeeks.org/functions-in-python/">https://www.geeksforgeeks.org/functions-in-python/</a>                                                                                                                                                              | 2 |

|     |        |                                                                                                                                                                                                                 |     |
|-----|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| 21. | Python | Exercise on List data type in Python:<br><a href="https://www.programiz.com/python-programming/list">https://www.programiz.com/python-programming/list</a>                                                      | 1   |
| 22. | Python | Exercise on Tuple data type in Python:<br><a href="https://www.programiz.com/python-programming/tuple">https://www.programiz.com/python-programming/tuple</a>                                                   | 1   |
| 23. | Python | Exercise on String data type in Python:<br><a href="https://www.programiz.com/python-programming/string">https://www.programiz.com/python-programming/string</a>                                                | 1   |
| 24. | Python | Exercise on Set data type in Python:<br><a href="https://www.programiz.com/python-programming/set">https://www.programiz.com/python-programming/set</a>                                                         | 1   |
| 25. | Python | Exercise on Dictionary data type in Python:<br><a href="https://www.programiz.com/python-programming/dictionary">https://www.programiz.com/python-programming/dictionary</a>                                    | 1   |
| 26. | Python | Exercise on Exception Handling in Python:<br><a href="https://www.programiz.com/python-programming/exception-handling">https://www.programiz.com/python-programming/exception-handling</a>                      | 2   |
| 27. | Python | Exercise on User defined Exception Handling in Python:<br><a href="https://www.programiz.com/python-programming/user-defined-exception">https://www.programiz.com/python-programming/user-defined-exception</a> | 2   |
| 28. | Numpy  | Exercise on Numpy create Array Using Python:<br><a href="https://www.w3schools.com/python/numpy_creating_arrays.asp">https://www.w3schools.com/python/numpy_creating_arrays.asp</a>                             | 3,4 |
| 29. | Numpy  | Exercise on Numpy Indexing in Array Using Python:<br><a href="https://www.w3schools.com/python/numpy_array_indexing.asp">https://www.w3schools.com/python/numpy_array_indexing.asp</a>                          | 3,4 |
| 30. | Numpy  | Exercise on Numpy Slicing in Array Using Python:<br><a href="https://www.w3schools.com/python/numpy_array_slicing.asp">https://www.w3schools.com/python/numpy_array_slicing.asp</a>                             | 3,4 |
| 31. | Numpy  | Exercise on Numpy Slicing in Array Using Python:<br><a href="https://www.w3schools.com/python/numpy_data_types.asp">https://www.w3schools.com/python/numpy_data_types.asp</a>                                   | 3,4 |
| 32. | Numpy  | Exercise on Numpy Array coping and viewing :<br><a href="https://www.w3schools.com/python/numpy_copy_vs_view.asp">https://www.w3schools.com/python/numpy_copy_vs_view.asp</a>                                   | 3,4 |
| 33. | Numpy  | Exercise on Numpy Array Shaping :<br><a href="https://www.w3schools.com/python/numpy_array_shape.asp">https://www.w3schools.com/python/numpy_array_shape.asp</a>                                                | 3,4 |
| 34. | Numpy  | Exercise on Numpy Array reshaping :<br><a href="https://www.w3schools.com/python/numpy_array_reshape.asp">https://www.w3schools.com/python/numpy_array_reshape.asp</a>                                          | 3,4 |
| 35. | Numpy  | Exercise on Numpy Array iteration:<br><a href="https://www.w3schools.com/python/numpy_array_iterating.asp">https://www.w3schools.com/python/numpy_array_iterating.asp</a>                                       | 3,4 |
| 36. | Numpy  | Exercise on Numpy Matrix joining<br><a href="https://www.w3schools.com/python/numpy_array_join_week_4.asp">https://www.w3schools.com/python/numpy_array_join_week_4.asp</a>                                     | 3,4 |
| 37. | Numpy  | Exercise on Numpy Array splitting<br><a href="https://www.w3schools.com/python/numpy_array_split.asp">https://www.w3schools.com/python/numpy_array_split.asp</a>                                                | 3,4 |
| 38. | Numpy  | Exercise on Numpy Array searching<br><a href="https://www.w3schools.com/python/numpy_array_search.asp">https://www.w3schools.com/python/numpy_array_search.asp</a>                                              | 3,4 |
| 39. | Numpy  | Exercise on Numpy Array sorting<br><a href="https://www.w3schools.com/python/numpy_array_sort.asp">https://www.w3schools.com/python/numpy_array_sort.asp</a>                                                    | 3,4 |
| 40. | Numpy  | Exercise on Numpy Array Random technique<br><a href="https://www.w3schools.com/python/numpy_random.asp">https://www.w3schools.com/python/numpy_random.asp</a>                                                   | 3,4 |
| 41. | Pandas | Exercise on Pandas basics:<br><a href="https://www.w3schools.com/python/pandas_tutorial.asp">https://www.w3schools.com/python/pandas_tutorial.asp</a>                                                           | 3,4 |
| 42. | Pandas | Exercise on Pandas installation:<br><a href="https://www.w3schools.com/python/pandas_getting_started.asp">https://www.w3schools.com/python/pandas_getting_started.asp</a>                                       | 3,4 |
| 43. | Pandas | Exercise on Pandas Series data<br><a href="https://www.w3schools.com/python/pandas_series.asp">https://www.w3schools.com/python/pandas_series.asp</a>                                                           | 3,4 |
| 44. | Pandas | Exercise on Pandas Data Frame:<br><a href="https://www.w3schools.com/python/pandas_dataframes.asp">https://www.w3schools.com/python/pandas_dataframes.asp</a>                                                   | 3,4 |
| 45. | Pandas | Exercise on Pandas Open CSV files:<br><a href="https://www.w3schools.com/python/pandas_csv.asp">https://www.w3schools.com/python/pandas_csv.asp</a>                                                             | 3,4 |

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| 46. | Pandas           | Exercise on Pandas Data analyzation:<br><a href="https://www.w3schools.com/python/pandas_analyzing.asp">https://www.w3schools.com/python/pandas_analyzing.asp</a>                                                                                                                                                                                                                                                                                                                                                   | 3,4       |
| 47. | Pandas           | Exercise on Pandas Data Cleaning techniques:<br><a href="https://www.w3schools.com/python/pandas_cleaning.asp">https://www.w3schools.com/python/pandas_cleaning.asp</a>                                                                                                                                                                                                                                                                                                                                             | 3,4       |
| 48. | Pandas           | Exercise on Pandas Data Correlation:<br><a href="https://www.w3schools.com/python/pandas_correlations.asp">https://www.w3schools.com/python/pandas_correlations.asp</a>                                                                                                                                                                                                                                                                                                                                             | 3,4       |
| 49. | Stats            | Perform Mean, Midian and mode:<br><a href="https://www.w3schools.com/python/python_ml_mean_median_mode.asp">https://www.w3schools.com/python/python_ml_mean_median_mode.asp</a>                                                                                                                                                                                                                                                                                                                                     | 2         |
| 50. | Stats            | Perform Standard Deviation:<br><a href="https://www.w3schools.com/python/python_ml_standard_deviation.asp">https://www.w3schools.com/python/python_ml_standard_deviation.asp</a>                                                                                                                                                                                                                                                                                                                                    | 2         |
| 51. | Machine Learning | Implement Linear Regression<br><a href="https://stackabuse.com/linear-regression-in-python-with-scikit-learn/">https://stackabuse.com/linear-regression-in-python-with-scikit-learn/</a>                                                                                                                                                                                                                                                                                                                            | 5         |
| 52. | Machine Learning | Perform Logistics Regression:<br><a href="https://towardsdatascience.com/logistic-regression-using-python-sklearn-numpy-mnist-handwriting-recognition-matplotlib-a6b31e2b166a">https://towardsdatascience.com/logistic-regression-using-python-sklearn-numpy-mnist-handwriting-recognition-matplotlib-a6b31e2b166a</a><br><a href="https://www.datacamp.com/community/tutorials/understanding-logistic-regression-python">https://www.datacamp.com/community/tutorials/understanding-logistic-regression-python</a> | 5         |
| 53. | Machine Learning | Exercise on Decision Tree:<br><a href="https://www.datacamp.com/community/tutorials/decision-tree-classification-python">https://www.datacamp.com/community/tutorials/decision-tree-classification-python</a>                                                                                                                                                                                                                                                                                                       | 6         |
| 54. | Machine Learning | Exercise on SVM:<br><a href="https://stackabuse.com/implementing-svm-and-kernel-svm-with-pythons-scikit-learn/">https://stackabuse.com/implementing-svm-and-kernel-svm-with-pythons-scikit-learn/</a>                                                                                                                                                                                                                                                                                                               | 6         |
| 55. | Machine Learning | Exercise on Time Series Analysis:<br><a href="https://www.dataquest.io/blog/tutorial-time-series-analysis-with-pandas">https://www.dataquest.io/blog/tutorial-time-series-analysis-with-pandas</a>                                                                                                                                                                                                                                                                                                                  | 6         |
| 56. | Machine Learning | Demonstration of Neural Networks:<br><a href="https://www.analyticsvidhya.com/blog/2020/07/neural-networks-from-scratch-in-python-and-r">https://www.analyticsvidhya.com/blog/2020/07/neural-networks-from-scratch-in-python-and-r</a>                                                                                                                                                                                                                                                                              | 7         |
| 57. | Deep Learning    | Exercise on MLP:<br><a href="https://machinelearningmastery.com/neural-networks-crash-course/">https://machinelearningmastery.com/neural-networks-crash-course/</a>                                                                                                                                                                                                                                                                                                                                                 | 7         |
| 58. | Deep Learning    | Exercise on Feed Forward neural networks:<br><a href="https://builtin.com/data-science/feedforward-neural-network-intro">https://builtin.com/data-science/feedforward-neural-network-intro</a>                                                                                                                                                                                                                                                                                                                      | 7         |
| 59. | Deep Learning    | Exercise on Neural Network:<br><a href="https://www.analyticsvidhya.com/blog/2019/08/detailed-guide-7-loss-functions-machine-learning-python-code/">https://www.analyticsvidhya.com/blog/2019/08/detailed-guide-7-loss-functions-machine-learning-python-code/</a>                                                                                                                                                                                                                                                  | 7         |
| 60. | Deep Learning    | Exercise on Linguistics using Machine learning in python:<br><a href="https://medium.com/towards-artificial-intelligence/natural-language-processing-nlp-with-python-tutorial-for-beginners-1f54e610a1a0">https://medium.com/towards-artificial-intelligence/natural-language-processing-nlp-with-python-tutorial-for-beginners-1f54e610a1a0</a>                                                                                                                                                                    | 7         |
| 61. | Deep Learning    | Text processing:<br><a href="https://pythonspot.com/category/nltk/">https://pythonspot.com/category/nltk/</a>                                                                                                                                                                                                                                                                                                                                                                                                       | 7         |
| 62. | Deep Learning    | Text Analysis<br><a href="https://medium.com/towards-artificial-intelligence/natural-language-processing-nlp-with-python-tutorial-for-beginners-1f54e610a1a0">https://medium.com/towards-artificial-intelligence/natural-language-processing-nlp-with-python-tutorial-for-beginners-1f54e610a1a0</a>                                                                                                                                                                                                                | 7         |
| 63. | Deep Learning    | Demonstrate Convolution Neural Network:<br><a href="https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53">https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53</a>                                                                                                                                                                                                                            | 7         |
| 64. | Deep Learning    | Perform CNN on CIFAR-10 Dataset<br>( <a href="https://www.analyticsvidhya.com/blog/2020/02/learn-image-classification-cnn-convolutional-neural-networks-3-datasets/">https://www.analyticsvidhya.com/blog/2020/02/learn-image-classification-cnn-convolutional-neural-networks-3-datasets/</a> )                                                                                                                                                                                                                    | 7         |
| 65. | Microsoft Azure  | Microsoft Azure Video Lectures at Microsoft Learning                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 10, 11,12 |